

KOGI

Independent Worker Operating System

Portfolio System

Updated: Shared Portfolios · Portfolio Collaboration · Crowdfunding Portable
Benefits as Portfolio Items · Collaboration Engine

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18 — Shared Portfolios & Portfolio Collaboration

A Shared Portfolio is a PortfolioSystem Portfolio component whose ComponentMetadata.owners[] contains two or more distinct user or organization entity IDs. All existing lifecycle, governance, CRDT, and analytics operations apply with the following multi-party extensions.

Shared Portfolio Types

Type	Owner Model	Governance
Team Portfolio	team entity	Admin-controlled; leader override
Organization Portfolio	organization entity	Multi-sig + governance proposal
Collective Portfolio	collective entity	Open contribution; light moderation
Cooperative Portfolio	cooperative entity	Full member vote required
Federation Portfolio	federation entity	Inter-org governance protocol
Crowdresourced Portfolio	open / community contributors	Contribution ledger; steward review + merge

Contribution Attribution Model

All contributions to a shared portfolio component are tracked as ContributionRecord objects appended to the component's EventLog:

```
ContributionRecord {
  record_id:          UUID,
  contributor_id:     EntityId,          // user | org | collective
  contribution_type:  ContributionType,  // Labor | Capital | Asset |
                                          // Knowledge | Artifact | Code | Design |

  Data
    portfolio_component_id: ComponentId,
    contribution_value:  ContributionValue { amount: Decimal, unit: String },
    attribution_weight:  f64,             // used in distribution calculations
    governance_status:  GovernanceStatus, // PendingReview | Accepted | Rejected |

  Merged
    linked_ledger_entry: Option<LedgerId>, // kogi-bank entry for capital
  contributions
}
```

Portfolio Collaboration Workflow States

State	Description
OPEN_CONTRIBUTION	Portfolio owner creates CrowdresourcingCampaign; specifies types, review policy, reward rules
SUBMITTED	Contributor submits labor, asset, capital, or knowledge contribution
STEWARD_REVIEW	Configurable gate: AutoAccept StewardReview GovernanceVote

State	Description
MERGED / ACCEPTED	ContributionRecord appended to EventLog; attribution weight updated
DISTRIBUTION_EVENT	Periodic or milestone-triggered; kogi-bank executes distribution to member accounts

Resource Sharing — GraphEdge Extension

Portfolio Resource Sharing is implemented as a ResourceShare edge in the GraphEdge system:

```
GraphEdge {
  edge_type: GraphEdgeType::ResourceShare,
  source_id: ComponentId,      // resource being shared
  target_id: EntityId,        // space | org | user receiving access
  metadata: {
    share_policy_id:          PolicyId,
    access_level:             Read | Fork | Contribute | Manage,
    expiry:                   Option<DateTime>,
    attribution_required: bool,
  }
}
```

Crowdresourcing Campaign Data Model

```
CrowdresourcingCampaign {
  campaign_id:                UUID,
  target_component_id:        ComponentId,
  contribution_types:         Vec<ContributionType>,
  open_to:                    Visibility, // Public | Space | Organization | Invite
  contribution_review:        ReviewPolicy, // AutoAccept | StewardReview | GovernanceVote
  contributor_rewards:         Vec<RewardRule>, // attribution_weight | equity | payment
  linked_bank_account:        Option<AccountId>,
}
```

CollaborationEngine Integration

- Compute real-time attribution weight updates for all contributors
- Detect merge conflicts between concurrent CRDT operations on shared components
- Score collaboration health: contributor diversity, contribution velocity, governance participation
- Generate Oba recommendations: "3 contributors have pending submissions awaiting steward review"
- Feed CollaborationScore into PortfolioHealth computational model

19 — Portable Benefits as Portfolio Items

Every portable benefit account is a PortfolioItem (kind: BenefitAccount) within the worker's personal portfolio, providing full lifecycle, governance, versioning, and analytics infrastructure.

```
BenefitAccount (PortfolioItem) {
  benefit_type:      BenefitType,  // Health | Dental | Vision | HSA |
                                // Retirement | PTO | Disability |
                                // ProfessionalDev | EmergencySavings |

  PortableSavings
    provider_id:      ProviderId,
    contribution_sources: Vec<ContributionSource>,
    balance:          Decimal,
    vesting_schedule: Option<VestingSchedule>,
    eligibility_rules: PolicySet,
    claim_workflow_id: Option<WorkflowId>,
    tax_treatment:    PreTax | PostTax | TaxFree,
    linked_bank_account: AccountId,  // kogi-bank BenefitAccount
}
```

BenefitType	Tax Treatment	Portfolio Analytics
Health (HSA)	Pre-tax	Coverage gap analysis, premium tracking, claim history
Retirement (SEP-IRA / 401k / PEP)	Pre-tax / Tax-deferred	Vesting progress, projected balance, contribution rate optimization
Paid Time Off	N/A	Accrual rate, usage, income replacement coverage
Disability / Workers' Comp	Post-tax (premiums)	Coverage adequacy vs. income, claim probability score
Professional Development	Post-tax	Skill development ROI, certification tracking, fund utilization
Portable Savings	Varies	Gig platform contribution tracking, balance projections, employer match equivalent